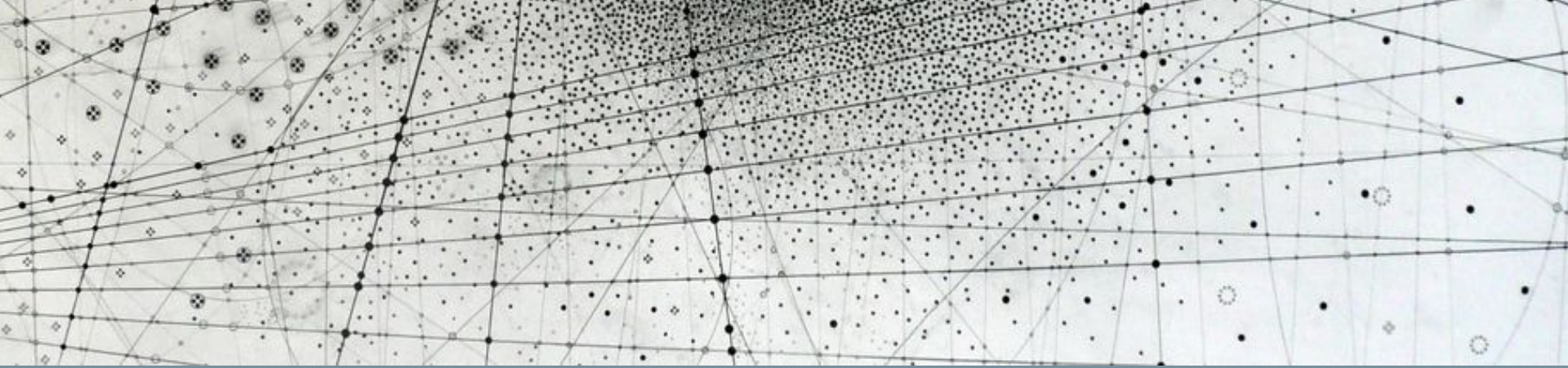




Research Methods in Machine Learning & Physics

Announcements

- Problem Set #3 posted last night
 - Due end-of-day next Wednesday the 26th
 - Topic: CLIP & SBI applied to astronomy data
- Next week: no class on Wednesday
- The following week (12/1 - 12/3): no class, but schedule 1-1 consultations
 - Look out for a Calendly link to schedule this
- Final project oral presentations on Monday 12/8 and Wednesday 12/10
 - Final project will count as 2 projects
 - 10+5 minute in-class presentation + 2-3 page extended abstract
- There will be 1 more Problem Set (#4), on the topic of gravitational waves, due before the end of the semester. I'll post this within the next ~week.

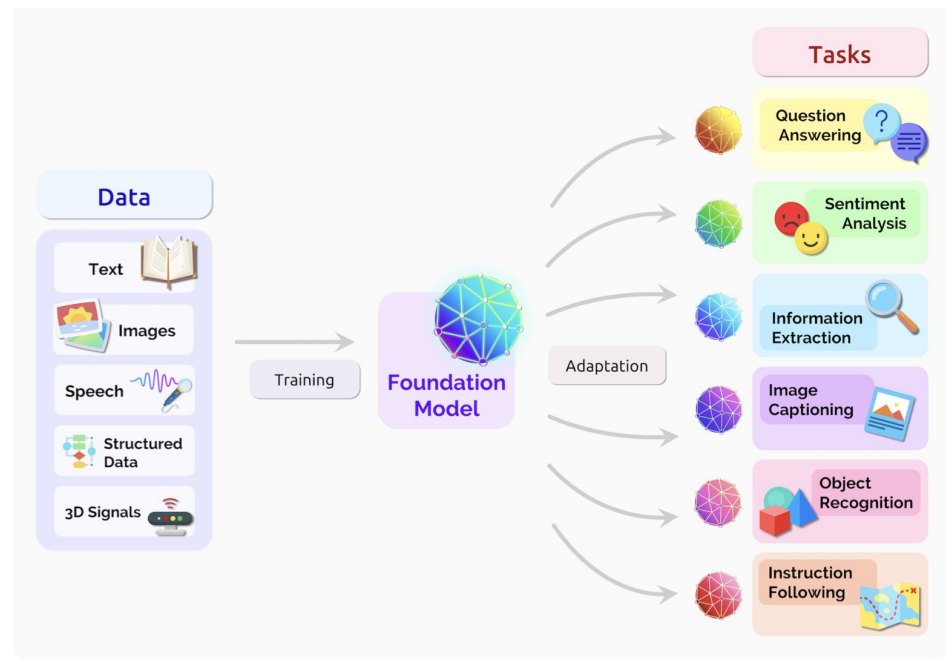
Quick recap

Foundation models are **"models trained on broad data (generally using self-supervision at scale) that can be adapted to a wide range of downstream tasks."**

On the Opportunities and Risks of Foundation Models

Rishi Bommasani* Drew A. Hudson Ehsan Adeli Russ Altman Simran Arora
Sydney von Arx Michael S. Bernstein Jeannette Bohg Antoine Bosselut Emma Brunskill
Erik Brynjolfsson Shyamal Buch Dallas Card Rodrigo Castellon Niladri Chatterji
Annie Chen Kathleen Creel Jared Quincy Davis Dorottya Demszky Chris Donahue
Moussa Doumbouya Esin Durmus Stefano Ermon John Etchemendy Kawin Ethayarajh
Li Fei-Fei Chelsea Finn Trevor Gale Lauren Gillespie Karan Goel Noah Goodman
Shelby Grossman Neel Guha Tatsunori Hashimoto Peter Henderson John Hewitt
Daniel E. Ho Jenny Hong Kyle Hsu Jing Huang Thomas Icard Saahil Jain
Dan Jurafsky Pratyusha Kalluri Siddharth Karamcheti Geoff Keeling Fereshte Khani
Omar Khattab Pang Wei Koh Mark Krass Ranjay Krishna Rohith Kuditipudi
Ananya Kumar Faisal Ladhak Mina Lee Tony Lee Jure Leskovec Isabelle Levent
Xiang Lisa Li Xuechen Li Tengyu Ma Ali Malik Christopher D. Manning
Suvar Mirchandani Eric Mitchell Zanele Munyikwa Suraj Nair Avaniika Narayan
Deepak Narayanan Ben Newman Allen Nie Juan Carlos Niebles Hamed Nilforoshan
Julian Nyarko Gray Ogut Laurel Orr Isabel Papadimitriou Joon Sung Park Chris Piech
Eva Portelance Christopher Potts Aditi Raghunathan Rob Reich Hongyu Ren
Frieda Rong Yusuf Roohani Camilo Ruiz Jack Ryan Christopher Ré Dorsa Sadigh
Shiori Sagawa Keshav Santhanam Andy Shih Krishnan Srinivasan Alex Tamkin
Rohan Taori Armin W. Thomas Florian Tramèr Rose E. Wang William Wang Bohan Wu
Jiajun Wu Yuhuai Wu Sang Michael Xie Michihiro Yasunaga Jiaxuan You Matei Zaharia
Michael Zhang Tianyi Zhang Xikun Zhang Yuhui Zhang Lucia Zheng Kaitlyn Zhou
Percy Liang*¹

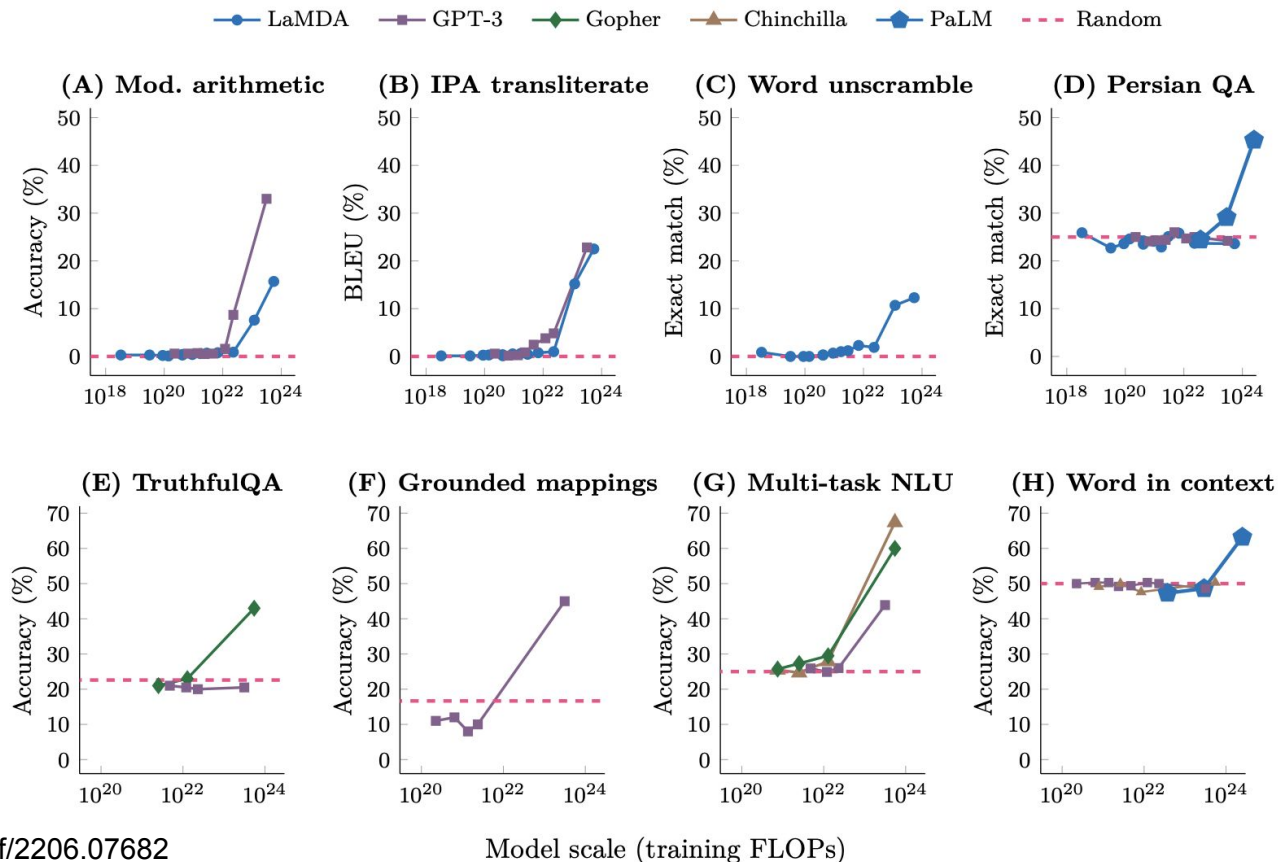
Center for Research on Foundation Models (CRFM)
Stanford Institute for Human-Centered Artificial Intelligence (HAI)
Stanford University



A slightly more relaxed definition of foundation models in science:

1. It has been pretrained on a large amount of data, with the **explicit intention of creating a rich latent representation** that serves as a foundation for other tasks;
2. It uses this latent representation to **finetune the model to a different downstream task** (be it the same type of task on a different dataset, or a different task altogether);
3. The **final performance on the downstream task improves** when the latent representation is used, compared to if the model is initialized without this representation.

Emergence of new skills at scale



Fine-tuning

Fine-tuning typically involves “nudging” your model in a new direction after the hefty “pre-training” stage – you don’t want to modify it too much, but just enough to adapt to a new setting.

e.g. “freeze” the weights of several layers (so they don’t change), but leave a few layers “unfrozen” so that they can adapt to new high-level features.

Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



In-context learning, i.e. learning without training!

The three settings we explore for in-context learning

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

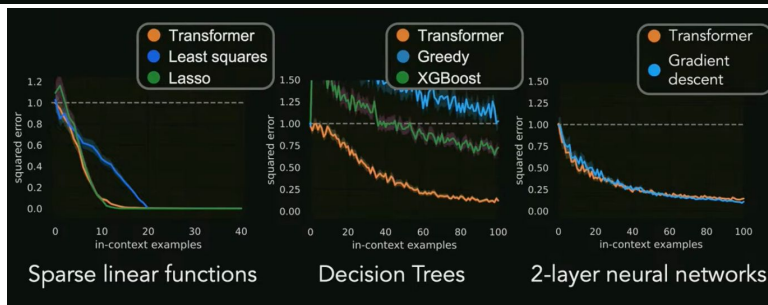
Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée
4 plush girafe => girafe peluche
5 cheese => ..... ← prompt
```

In-Context Learning: Learning *without* weight updates

$9 - 2 = 11$, $13 - 7 = 20$, $2 - 5 = 7$, $4 - 1 = ??$



How is this possible?? → approximating gradient descent:

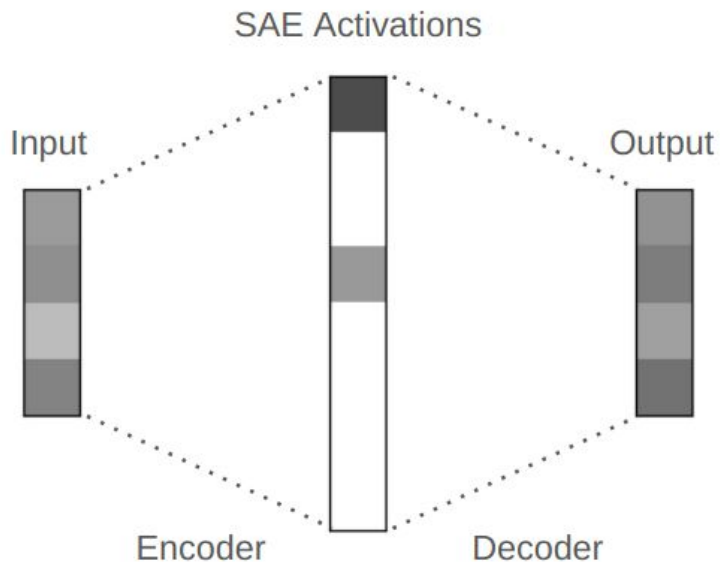
Transformers Learn In-Context by Gradient Descent

Why Can GPT Learn In-Context?
Language Models Implicitly Perform Gradient Descent as
Meta-Optimizers

2. Linear self-attention *can* emulate gradient descent on a linear regression task

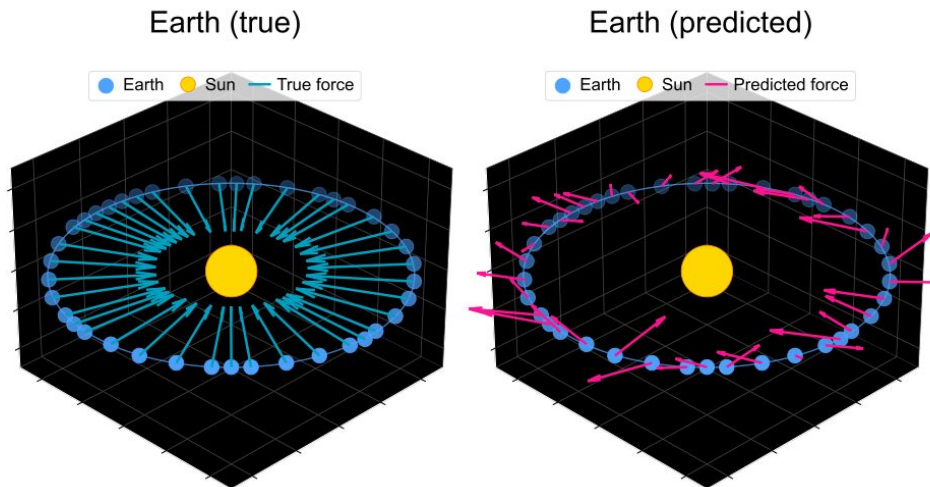
Interpretability

Sparse Autoencoders (SAEs) can reveal important features



What Has a Foundation Model Found? Using Inductive Bias to Probe for World Models

Keyon Vafa¹ Peter G. Chang² Ashesh Rambachan² Sendhil Mullainathan²

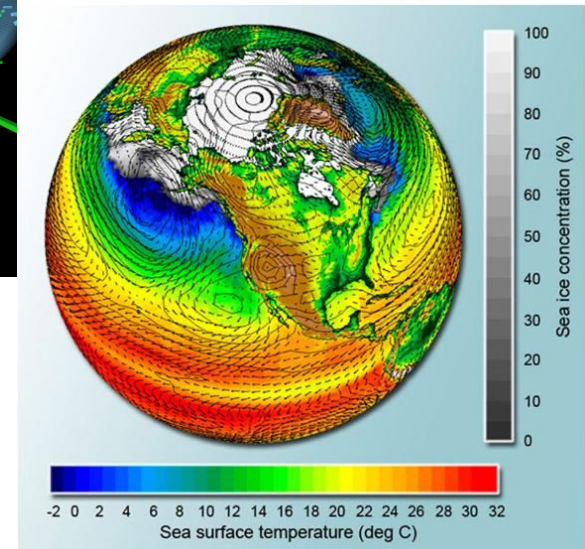
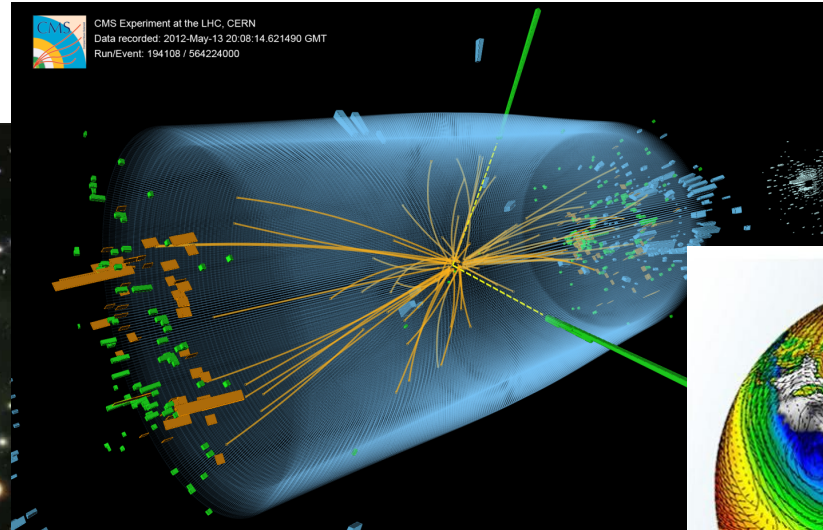
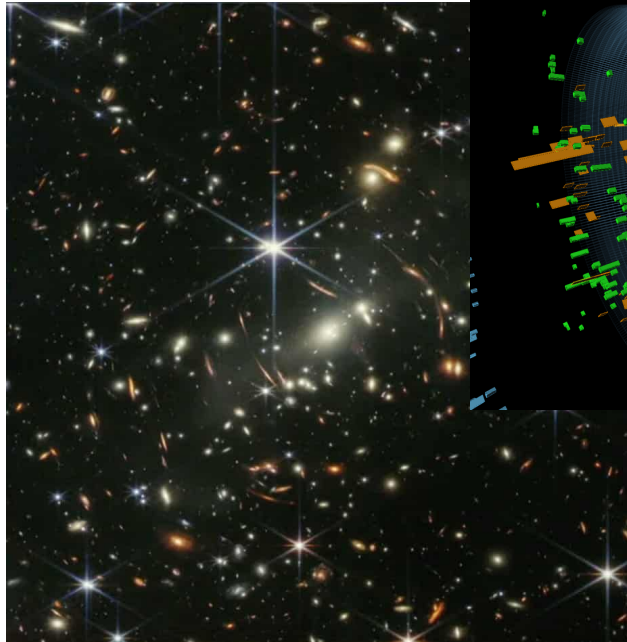


Foundation Models, Part II

Where in physics do we expect to find foundation models being used?

- Data-rich
- Multi-modal / highly specialized data
 - Existing foundation models are insufficient to describe the data
- Already comfortable with high-performance computing work
- Unsolved yet easily-verifiable tasks
- Generic or nearly-universal phenomena
 - e.g. objects in space that can be measured with multiple telescopes, identical fundamental particles, fluid dynamics across a variety of scales
- Funding and societal preferences/priorities

Where in physics do we expect to find foundation models being used?



"AI scientists"



Building the AI Physicist

FirstPrinciples is building the AI Physicist, a domain-specialized system designed to understand the nature of reality.

[Learn More ↗](#)

The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery

Chris Lu^{1,2,*}, Cong Lu^{3,4,*}, Robert Tjarko Lange^{1,*}, Jakob Foerster^{2,†}, Jeff Clune^{3,4,5,†} and David Ha^{1,†}

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sakana.ai

The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery

August 13, 2024



periodic labs

ACCELERATE SCIENCE

Today, we introduce Periodic Labs. Our goal is to create an AI scientist.

Kosmos: An AI Scientist for Autonomous Discovery

Ludovico Mitchener^{*,1,†}, Angela Yiu^{*,1}, Benjamin Chang^{*,1,2}, Mathieu Bourdenx^{3,4,5}, Tyler Nadolski¹, Arvis Sulovari¹, Eric C. Landsness^{5,6}, Dániel L. Barabási^{7,8}, Siddharth Narayanan¹, Nicky Evans⁹, Shriya Reddy¹⁰, Martha Foiani^{3,4}, Aizad Kamal⁶, Leah P. Shriver^{11,12,13}, Fang Cao¹⁰, Asmamaw T. Wassie¹, Jon M. Laurent¹, Edwin Melville-Green¹, Mayk Caldas¹, Albert Bou¹, Kaleigh F. Roberts¹⁴, Sladjana Zagorac¹⁵, Timothy C. Orr⁶, Miranda E. Orr^{6,16}, Kevin J. Zwezdaryk^{17,18,19}, Ali E. Ghareeb¹, Laurie McCoy¹, Bruna Gomes¹⁰, Euan A. Ashley¹⁰, Karen E. Duff^{3,4,5}, Tonio Buonassisi^{9,20}, Tom Rainforth², Randall J. Bateman^{5,6}, Michael Skarlinski¹, Samuel G. Rodrigues^{1,7,‡}, Michaela M. Hinks^{1,†}, Andrew D. White^{1,7,‡}

<https://arxiv.org/pdf/2511.02824>

What are physicists for?

- Making discoveries
- Collecting data
- Analyzing data
- Writing papers
- Giving talks
- Mentoring / teaching students

What does it mean to “accelerate science”?

Is this something we should aim for?

Why should human scientists exist?

In-class work: Paper Discussion

Summarize the paper's overall message, methods, and results.

What are its strengths and weaknesses?

Task #1: A review of Foundation Models in HEP – <https://arxiv.org/pdf/2509.21434>

Task #2: A recent Foundation Model for PDEs – <https://arxiv.org/pdf/2509.13805>

Task #3: A recent Foundation Model for Astrophysics – <https://arxiv.org/pdf/2510.17960>